

CSE343/CSE543/ECE363/ECE563
Machine Learning (UG/PG)
Winter 2025

Mid-Semester Examination (16 points)

Roll number:

Duration: 1 hour

Instructions

- **Institute's Academic Dishonesty Policy is Applicable.**
- All questions are compulsory.
- There can be multiple ways to approach a question. Justify all your answers appropriately. Questions without justification will get zero marks.
- Students will not be allowed to use the washroom during the test. Any request in this regard will not be entertained.
- Any student trying to copy during the test will be awarded a grade of F in the course along with his accomplice.

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1. **[CO1 CO2] Logistic Regression** **(6 points)**
Recall the setup of logistic regression: We assume that the posterior probability is of the form

$$p(Y = 1|\mathbf{x}) = \frac{1}{1 + e^{-\beta^T \mathbf{x}}} \quad (1)$$

This assumes that $Y|\mathbf{X}$ is a Bernoulli random variable. We now turn to the case where $Y|\mathbf{X}$ is a multinomial random variable over K outcomes. This is called softmax regression, because the posterior probability is of the form

$$p(Y = k|\mathbf{x}) = \mu_k(\mathbf{x}) = \frac{e^{\beta_k^T \mathbf{x}}}{\sum_{j=1}^K e^{\beta_j^T \mathbf{x}}}, \quad (2)$$

which is called the softmax function. Assume we have observed data $D = \{\mathbf{x}_i, y_i\}_{i=1}^N$. Our goal is to learn the weight vectors $\beta_1, \dots, \beta_K$.

(a) **[2 pts]** Find the negative log likelihood of the data $l(\beta_1, \dots, \beta_K)$.

(b) **[2 pts]** We want to minimize the negative log likelihood. To combat overfitting, we put a regularizer on the objective function. Find the gradient w.r.t. β_k of the regularized objective

$$l(\beta_1, \dots, \beta_K) + \lambda \sum_{k=1}^K \|\beta_k\|^2 \quad (3)$$

(c) **[2 pts]** State the gradient updates for both batch gradient descent and stochastic gradient descent.

2. **[CO1, CO2, CO4] Short answer type questions** **(6 points)**

- (a) State some relations between autoencoders and PCA (1 point)
- (b) What is the primary purpose of the pooling layer in a CNN? (1 point)
- (c) Explain the concept of a hyperplane in SVM and its role in classification. (1 point)
- (d) Suppose there are total n data points in a linearly separable dataset and you are training a hard-margin SVM binary classifier, where k data points belong to the positive class, and m data points belong to the negative class, and then what should a maximum number of support vectors form a maximum margin classifier? Justify your answer using an example. (1 point)

- (e) In a linear hard margin SVM, if you slightly move (perturb) a training sample that is not a support vector, does it affect the learned decision boundary or not? Justify your answer (1 point)
- (f) Which regularization method leads to weight sparsity? Explain why. (1 point)

3. [CO1, CO2] **Practical ML**

(2 points)

You are training a ResNet-18 model, a CNN-based architecture for image classification on the CIFAR-10 dataset. You initialized the model with 3 different initialization states and with 3 different types of hyper-parameters. Based on the training-validation loss vs epoch curves, you have to tell about the model if it is a Good Fit, Underfit, or Overfit. Also, comment about the bias and variance in the model training process. The loss function used for training is Cross Entropy Loss.

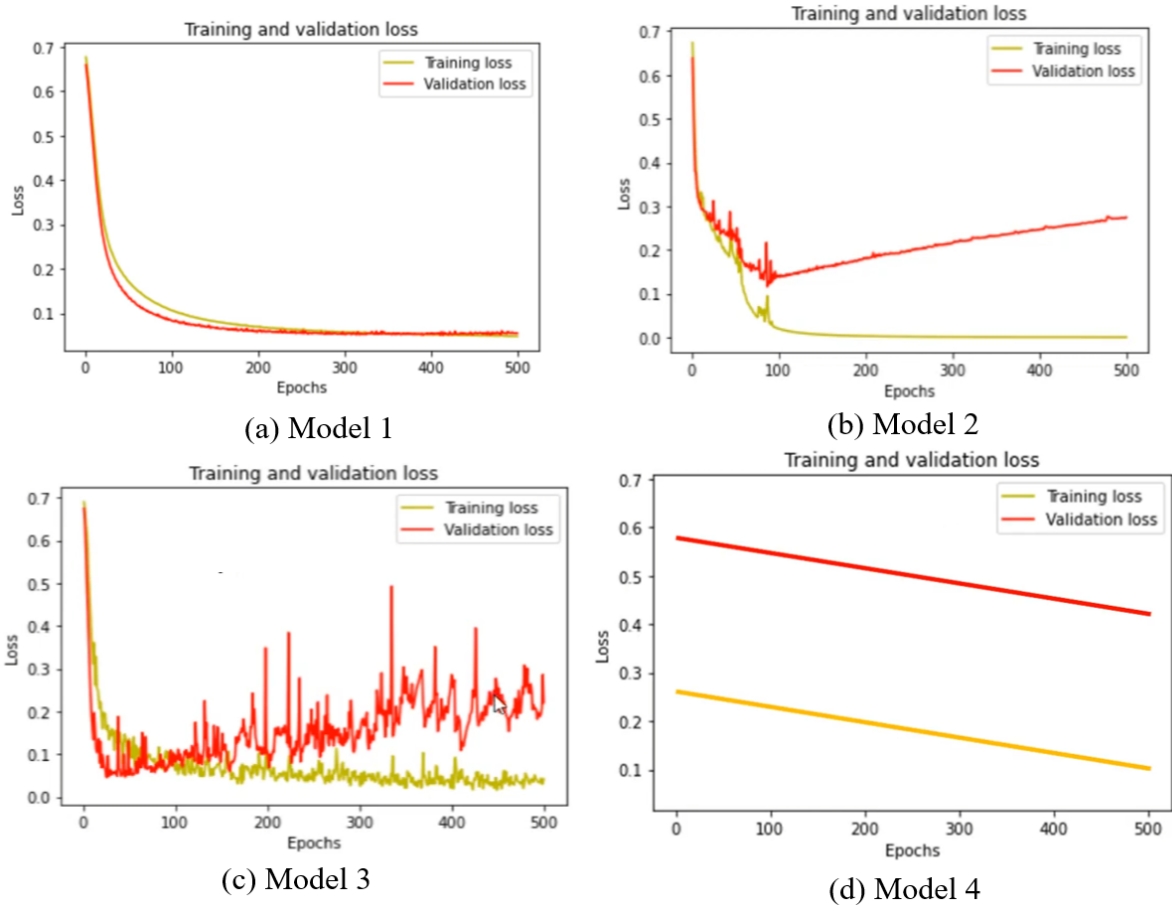


Figure 1: Train and Validation Loss vs Epoch Curves

4. [CO2] **Convolutional Neural Network**

(2 points)

Consider a CNN with two convolutional layers followed by a fully connected network for classification. **Input:** The input size will always remain $64 \times 64 \times 3$. The architecture is as follows:

Layer 1 (Conv1): Filters: 32, Kernel: 5×5 , Stride: 1, Padding: 0, Dilation: 2

Layer 2 (Conv2): Filters: 64, Kernel: 3×3 , Stride: 2, Padding: 0, Dilation: 1

Fully Connected Layers: FC1: 128 neurons (ReLU), FC2: 10 neurons (Softmax)

Compute the output size of the feature map after both convolutional layers and determine the total number of trainable parameters.